# Homework 3, Sammy Abdella, sn3136

## 1.

**1(a)** **No it's not valid**

**1(b)** The argument doesnt have a valid form *affirming the consequent*. A counterexample is $p = F$, $q = T$ then $p \to q$ is true and $q$ is true but $p$ is false

**1(c)** truth table for $((p \to q) \wedge q) \to p$:

| $p$ | $q$ | $(p \to q)$ | $(p \to q) \wedge q$ | $((p \to q) \wedge q) \to p$ |
|---|---|---|---|---|
| $T$ | $T$ | $T$ | $T$ | $T$ |
| $T$ | $F$ | $F$ | $F$ | $T$ |
| $F$ | $T$ | $T$ | $T$ | $F$ |
| $F$ | $F$ | $T$ | $F$ | $T$ |

since the final column is *false* when $p = F$ and $q = T$ then the formula is not a tautology which confirms the rule is invalid

## 2.

| | | |
|---|---|---|
| 1. | $rainy \to (cold \vee windy)$ | Premise |
| 2. | $\neg cold$ | Premise |
| 3. | $windy \to uncomfortable$ | Premise |
| 4. | $\neg uncomfortable$ | Premise |
| 5. | $stormy \to rainy$ | Premise |
| 6. | Assume $stormy$ | assumption (for contradiction) |
| 7. | $rainy$ | $5, 6$ Modus Ponens |
| 8. | $cold \vee windy$ | $1, 7$ Modus Ponens |
| 9. | $windy$ | $8, 2$ Disjunctive Syllogism |
| 10. | $uncomfortable$ | $3, 9$ Modus Ponens |
| 11. | $\perp$ | $10, 4$ Contradiction |
| 12. | $\neg stormy$ | 6–11 negation introduction (so indirect proof) |

## 3.

**3(1)** $\forall x\Big((novel(x) \wedge bestselling(x)) \to (engaging(x) \wedge wellwritten(x))\Big)$.

**3(2)** $\exists x\Big(textbook(x) \wedge \neg difficult(x)\Big)$.

**3(3)** $\Big(\forall x\,(biography(x) \to informative(x))\Big) \;\wedge\; \Big(\exists x\,(novel(x) \wedge bestselling(x))\Big)$.

## 4.

**4(a)** $\forall x\,\exists y\,(comedy(y) \wedge enjoys(x, y))$.

**4(b)** $\exists y\,(drama(y) \wedge \forall x\, enjoys(x, y))$.

**4(c)** $\forall x\,\exists y\,\big(comedy(y) \wedge \neg enjoys(x,y)\big)$.

**4(d)** $\forall x\,\forall y\,\big(directed(x,y) \rightarrow enjoys(x,y)\big)$.

**4(e)** $\forall y\,\big(acclaimed(y) \rightarrow \exists x\, directed(x,y)\big)$.

**5.**

**5(a)** $\neg(\forall x\,(P(x) \rightarrow Q(x))) \equiv \exists x\,(P(x) \wedge \neg Q(x))$.

**5(b)** $\neg(\exists x\,(R(x) \wedge \forall y\, S(x,y))) \equiv \forall x\,(\neg R(x) \vee \exists y\, \neg S(x,y))$.

**5(c)** $\neg(\forall x\,(\exists y\, P(x,y) \rightarrow \forall z\, Q(z))) \equiv \exists x\,(\exists y\, P(x,y) \wedge \exists z\, \neg Q(z))$.